

MODULE-II: Green Chemistry

Green chemistry is also called sustainable chemistry, is an area of chemistry and chemical engineering focused on designing of products and processes that minimize or eliminate the use and generation of hazardous substances to humans, animals, plants and the environment.

Principles of Green Chemistry

Paul Anastas (who directed the green chemistry program at the USEPA) and John C. Warner published a set of principles to guide the practice of green chemistry.

In green chemistry use of safe and environmentally designed substances including solvents wherever possible. The design of energy efficient process avoid the product waste.

The 12 principles of green chemistry are:

1. Pollution prevention
2. Atom Economy
3. Less Hazardous chemical substances
4. Design safer chemicals
5. Use renewable feedstocks
6. Safer solvents and auxiliaries

7. Design for energy efficiency
8. Reduce derivatives
9. Catalysis
10. Design for Degradation
11. Real time analysis for pollution prevention
12. Inherently safer chemistry for accident prevention.

1. Pollution Prevention:

Preventing waste is better than treatment or cleaning up waste after it is created.

2. Atom Economy:

Synthetic method should try to maximize the incorporation of all material used in the process of final product. This means use waste will be generated as a result.

3. Less Hazardous Chemical Substances:

Synthetic method should avoid using or generating toxic substances to human and environment.

4. Design Safer Chemicals:

Chemical products should be designed to achieve their desired function while being as non-toxic as possible.

5. Safer Solvents & Auxillaries:

The use of safer solvents and auxillaries can help minimize the potential for chemical accidents such as explosions and fire. It can also help to

reduce toxic waste and improve the sustainability of synthetic protocols.

6. Design For Energy Efficiency:

Energy requirement should be minimized, process should be conducted at ambient temperature and pressure wherever possible.

7. Use Renewable Feedstocks:

Renewable Feedstocks can be used as a raw material or a full source. In manufacturing for a wide range of products, it also helps to reduce environmental impact, production cost.

8. Reduce Derivatives:

Unnecessary generation of derivatives such as the protecting groups should be minimized or avoided if possible.

9. Catalysis:

Catalytic reagents that can be used in small quantity to repeat a reaction are superior to stoichiometric reagents (consumed in a reaction).

10. Design for Degradation:

Chemical products should be designed so that they do not pollute the environment when their function is complete, they should be broken down into non-harmful products.

11. Real Time Analysis for Pollution Prevention:

Analytical methodology need to be further developed to permit real time, in process of monitoring and control before hazardous substance form.

12. Inherently Safer Chemistry for Accident Prevention:

Whenever possible the substances in a process and the forms of those substances should be chosen to minimize the risk such as explosions, fire and accidental injuries.

Green Solvent

The solvents are consumed in large quantities in many chemical synthesis as well as for cleaning and degreasing. Traditional solvents are often toxic or chlorinated.

Green solvents are generally derived from renewable resources and biodegrade, natural occurring products.